



# McAfee Labs Threat Advisory

## Ransom-Ryuk

October 29, 2020

McAfee Labs periodically publishes Threat Advisories to provide customers with a detailed analysis of prevalent malware. This Threat Advisory contains behavioral information, characteristics, and symptoms that may be used to mitigate or discover this threat, and suggestions for mitigation in addition to the coverage provided by the DATs.

To receive a notification when a Threat Advisory is published by McAfee Labs, select to receive “Malware and Threat Reports” at the following URL: <https://www.mcafee.com/enterprise/en-us/sns/preferences/sns-form.html>.

### Summary

Ransom-Ryuk is a family of ransomware that, on execution, encrypts files present on the user’s system. The compromised user must pay the attacker with a ransom to get the files decrypted.

Although traditionally ransomware has been known to be distributed via Exploit Kits (EK) and malicious email campaigns, Ransom-Ryuk is suspected to be distributed via targeted attacks. Attackers may already have access to an organization’s network via prior successful hacking/infection attempts.

Detailed information about the threat, its propagation, characteristics, and mitigation are in the following sections:

- [Infection and Propagation Vectors](#)
- [Mitigation](#)
- [Characteristics and Symptoms](#)
- [Restart Mechanism](#)
- [Access Protection Rules](#)
- [McAfee Foundstone Services](#)

We recommend following the Mitigation section listed above and creating the Expert rules which will provide an indication of compromise.

The Characteristics and Symptoms section also contains detection names which may indicate that a Ransom Ryuk attack is imminent. If these are discovered in your environment, we recommend following the steps in the Mitigation section and performing a full ODS scan with process scanning enabled.

The minimum DAT versions required for detection are:

| Detection Name                | DAT Version | Date       |
|-------------------------------|-------------|------------|
| Ransom-Ryuk!<PartialHash>     | V2: 9788    | 10/28/2020 |
| Real Protect-SS!<PartialHash> | V3: 4240    |            |
| Real Protect-LS!<PartialHash> |             |            |
| Real Protect-EC!<PartialHash> |             |            |

Table 1: Minimum DAT versions for coverage.

The Threat Intelligence Library contains the date that the above signatures were most recently updated. Please review the [Threat Intelligence Library](#) for the most up-to-date coverage information.

## Infection and Propagation Vectors

- Currently, the infection vector of Ransom-Ryuk is unknown.
- Most ransomware campaigns typically spread via Exploit Kits and malspam campaigns instrumented via various botnets.
- Ransom-Ryuk is however suspected to be distributed using highly targeted attacks such as brute forcing of RDP connections on unprotected systems in an organization's network and malicious use of Cobalt Strike.
- Once the attackers have access to the organization's network, Ransom-Ryuk may be deployed to business-critical system to cause maximum disruption of services and in-turn warrant a considerable ransom.

## Mitigation

Mitigating the threat at multiple levels such as file, registry, and URL can be achieved at various layers of McAfee products. Browse the product guidelines available [here](#) to mitigate the threats based on the behavior described below in the [Characteristics and symptoms](#) section.

Never open unsolicited emails and their attachments. Also, be wary of suspicious looking advertisements. Customers are advised to regularly update their infrastructure (both operating system and application software) with the latest patches to ensure full coverage in addition to updated McAfee Anti-Virus software.

## McAfee EDR

As APTs and nation states increase the use of ransomware payloads and malware continues to behave more like red teamers, organizations must increase their visibility and detection capabilities across their endpoints. McAfee MVISION EDR provides visibility, behavioural detection and triage capabilities across ransomware attack tactics and techniques. In particular, MVISION EDR provides visibility and detection for key MITRE ATT&CK techniques used by Ryuk, in particular:

- **TA 0002 - Execution:**
  - T1059 – Command and Scripting Interpreter
- **TA 0003 - Persistence:**
  - T1547 – Boot or Logon AutoStart Execution
- **TA 0004 - Privilege Escalation:**
  - T1547 – Boot or Logon AutoStart Execution
  - T1055 – Process Injection
- **TA 0005 - Defence Evasion:**
  - T1562 – Impair Defences
  - T1055 – Process Injection
- **TA 0007 - Discovery:**
  - T1083 – File and Directory Discovery
  - T1016 – System Network Configuration Discovery
- **TA 0040 - Impact:**
  - T1489 – Service Stop

MSIVISION EDR also provides visibility and behavioural detection for other steps that are common to these types of campaigns, including initial access, credential access, lateral movement, command and control and exfiltration, among others.

## McAfee Endpoint Security

Mitigation methods for assorted malware is available in the following product guide. Any specific mitigation steps, if necessary, would be described later in this advisory:

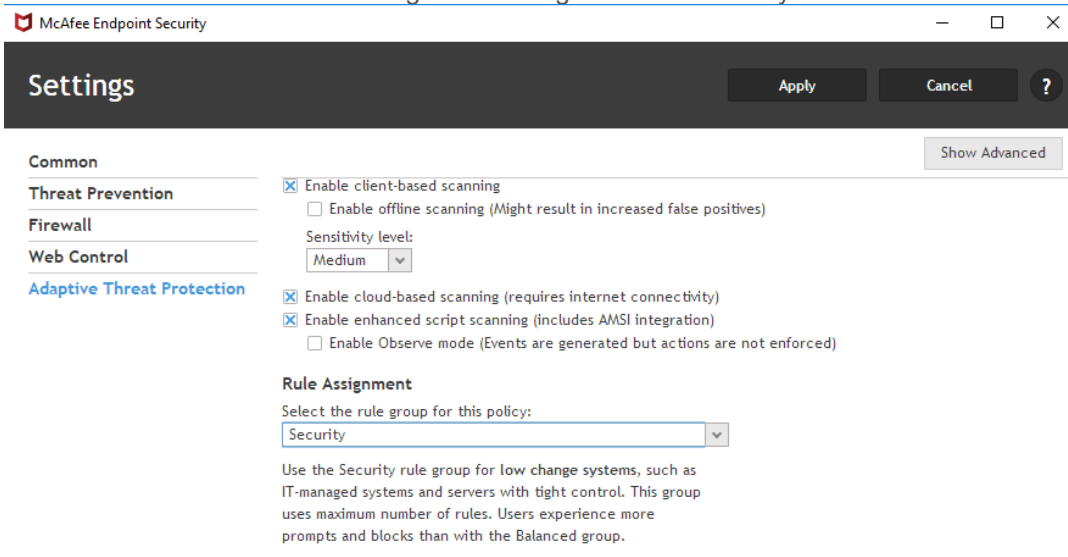
[http://b2b-download.mcafee.com/products/evaluation/Endpoint\\_Security/Evaluation/ens\\_1000\\_help\\_0-00\\_en-us.pdf](http://b2b-download.mcafee.com/products/evaluation/Endpoint_Security/Evaluation/ens_1000_help_0-00_en-us.pdf)

## ePolicy Orchestrator (ePO)

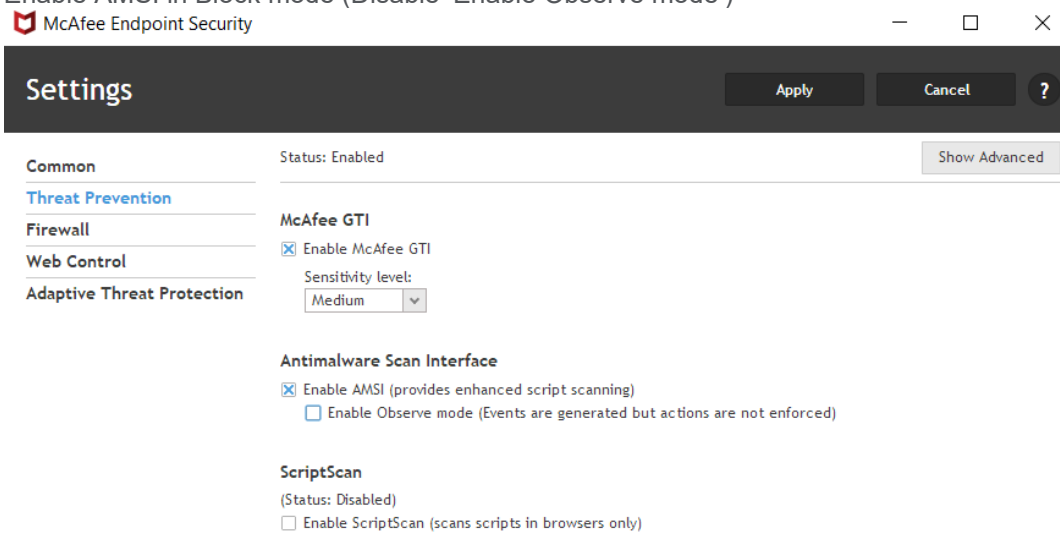
- To block the access to USB drives through ePO DLP policy, refer to this [tutorial](#).

## Endpoint Security 10.x

- Refer to article [KB86577](#) to create an Endpoint Security Threat Prevention user-defined Access Protection Rule for a file or folder registry.
- [ENS ATP Ccustomers should change Rule Assignment to 'Security'](#)



- Perform Full ODS scan with Process Scanning Enabled
- Enable AMSI in Block mode (Disable 'Enable Observe mode')



- Custom Expert Rules (*NOTE: These will not prevent Encryption, but will trigger Alerts in the product and EPO which can be an indicator of compromise*):

### Rule: 1 [Blocks the creation of .ryk files]

```
Rule {
  Process {
    Include OBJECT_NAME { -v *.exe }
  }
  Target {
    Match FILE {
      Include OBJECT_NAME { -v "*.ryk" }
      Include -access "CREATE"
    }
  }
}
```

### Rule: 2 [Blocks the execution of process with 8 LAN command line]

```
Rule {
  Process {
    Include OBJECT_NAME { -v *.exe }
```

```

}
Target {
Match PROCESS {
Include PROCESS_CMD_LINE { -v "***8 LAN" }
Include -access "CREATE"
}
}
}
}

```

#### Rule: 3 [Blocks the creation of pipe associated with Cobalt Strike]

```

Rule {
Process {
Include OBJECT_NAME { -v *.exe }
}
Target {
Match FILE {
Include OBJECT_NAME { -v "***pipe\MSSE*" }
Include -access "CONNECT_NAMED_PIPE"
}
}
}
}

```

#### Rule: 4 [Blocks the creation of pipe associated with Cobalt Strike]

```

Rule {
Process {
Include OBJECT_NAME { -v *.exe }
}
Target {
Match FILE { Include OBJECT_NAME { -v "***pipe\status_*" }
Include -access "CONNECT_NAMED_PIPE"
}
}
}
}

```

- Exploit Prevention

Enable Rule 6153 in Block and Report

Common

Threat Prevention

Options

Access Protection

Exploit Prevention

On-Access Scan

On-Demand Scan

Status: Enabled

Hide Advanced

Delete Add Expert Rule

| ID   | Name                                      | S...  | Block                               | Report                              | St... | Type  | Origin   |
|------|-------------------------------------------|-------|-------------------------------------|-------------------------------------|-------|-------|----------|
| 6153 | Malware Behavior: Ryuk Ransomware acti... | Di... | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | E...  | Files | McAfee-c |

- Access Protection Rule

| Rule# | Action                | Windows 6.x | File Actions to Prevent |
|-------|-----------------------|-------------|-------------------------|
| 1     | SubRule Type: Files   |             | Create                  |
|       | Executables [Include] | *           |                         |
|       | SubRule [Include]     | *.RYK       |                         |

## VirusScan Enterprise

- Refer to article [KB53346](#) to use Access Protection policies in VirusScan Enterprise to protect against viruses that can disable regedit.
- Refer to article [KB53355](#) to use Access Protection policies in VirusScan Enterprise to protect against viruses that can disable Task Manager.
- Refer to article [KB53356](#) to use Access Protection policies in VirusScan Enterprise to prevent malware from changing folder options.
- Perform Full ODS scan with Process Scanning Enabled
- Access Protection Rules

| Rule# | Action                       | Windows 6.x | File Actions to Prevent |
|-------|------------------------------|-------------|-------------------------|
| 1     | File/Folder Blocking Rule    |             | New files being created |
|       | Processes to include         | *           |                         |
|       | File or Folder Name to Block | *.RYK       |                         |

## Host Intrusion Prevention

- To block applications using a Host Intrusion Prevention custom signature, refer to article [KB71329](#).
- To create an application blocking rules policies to prevent the binary from running, refer to article [KB71794](#).
- To create an application blocking rules policies that prevents a specific executable from hooking any other executable, refer to article [KB71794](#).

## Others

- To disable the Autorun feature on Windows remotely using Windows Group Policies, refer to this [article](#) from Microsoft.

## Characteristics and Symptoms

### Ransom-Ryuk Dropper

Ransom-Ryuk begins its infection on an endpoint by execution of a dropper binary. The dropper consists of two additional second-stage binaries which are x86 and x64 versions of the same binary.

On execution, the dropper carries out the following sequence of steps:

- Checks the operating system (OS) version against a hard-coded value representing Microsoft Windows version 5. That is, it checks to verify if the current OS of the endpoint is Windows XP/Windows 2000.
- The dropper also checks the architecture of the OS (checks if the OS is 32bit or 64bit).
- Depending on the architecture of the OS, the dropper writes the second-stage binary into a location on the system depending on the OS version:  
For Windows 7 and above: C:\users\Public\<random\_characters>.exe  
For Windows XP/Windows 2000: C:\Documents and Settings\Default User\<random\_characters>.exe  
Other variants have been observed to drop copies of itself into the execution location.
- Once the second-stage binary is dropped on the filesystem, it is executed by the dropper process using the ShellExecute() API.

### Second-Stage Binary

The second-stage binary may or may not use the path to the original dropper as an argument in its command-line parameters. If the path to the dropper is present in the command line to the second-stage binary, the dropper binary on disk is deleted by the second-stage binary process.

On execution, the second-stage binary carries out the following sequence of steps:

- Persistence: Setups registry persistence for the second-stage binary using: cmd.exe /C REG ADD command. More details are in the Persistence section below.
- Privileges: The malware now continues to acquire the "SeDebugPrivilege" on the endpoint to elevate its current privileges.
- Process Enumeration: The second-stage binary enumerates the list of processes currently running on the system to build a list of process that are not run under the NT AUTHORITY system account.
- Process selection: This list of processes is further refined based on the criteria that the process name must NOT belong to any of the process names:

```
csrss.exe
explore.exe
lsaas.exe
```

- **Process Injection:** Process injection on the selected list of selected processes is carried out using a regular process injection technique where memory is allocated to the remote process, malicious code is written to the process and then the malicious code is executed via a remote thread in the target process.

### Injected Code

The second-stage binary injects a copy of itself into a target process and starts a new thread to continue its malicious activities. This code is responsible for encryption of files on all the drives on the system. The ransomware skips encryption of files or directories containing the following keywords:

- Windows
- Mozilla
- \$RecycleBin
- Chrome
- AhnLab

Ransom-Ryuk also has the capability to encrypt files on network shares. This done by enumerating files on each available network resource and performing subsequent encryption of files.

Ryuk Ransomware use "icacls.exe" to get full control over mentioned folder and all its subfolders.

- `Icacls "C:\*" /grant Everyone:F /T /C /Q`  
F gives Full Access and /T /C /Q applies the permissions to subfolders.

### Encryption Scheme

Ransom-Ryuk uses three sets of keys on the endpoint:

- **3rd level AES (symmetric):** Used to encrypt file contents on disk.
- **2nd level RSA key Pair:** Used to encrypt the 3rd level AES key and append to end of encrypted user files. The private key pair here is pre-encrypted with the 1st level RSA public key.
- **1st level RSA Key Pair:** 1<sup>st</sup> level private key used to decrypt the 2nd level private key once ransom is paid.

### Ransom Notes

Ransom notes are generated in each directory that has been encrypted by the ransomware. These files are usually named "RyukReadMe.txt"

Your network has been penetrated.

All files on each host in the network have been encrypted with a strong algorithm.

Backups were either encrypted or deleted or backup disks were formatted.

Shadow copies also removed, so F8 or any other methods may damage encrypted data but not recover.

We exclusively have decryption software for your situation

No decryption software is available in the public.

DO NOT RESET OR SHUTDOWN - files may be damaged.

DO NOT RENAME OR MOVE the encrypted and readme files.

DO NOT DELETE readme files.

This may lead to the impossibility of recovery of the certain files.

To get info (decrypt your files) contact us at

WayneEvenson@protonmail.com

or

WayneEvenson@tutanota.com

BTC wallet:

[REDACTED]

Ryuk

No system is safe

Figure: Ransom-Ryuk Ransom Note

Some versions of RYUK drops an HTML note (RyukReadMe.html) instead of the text file as ransomnote. This HTML contains just minimum information, such as the email address, the name RYUK and a message at the bottom (balance of shadow universe) shown below:

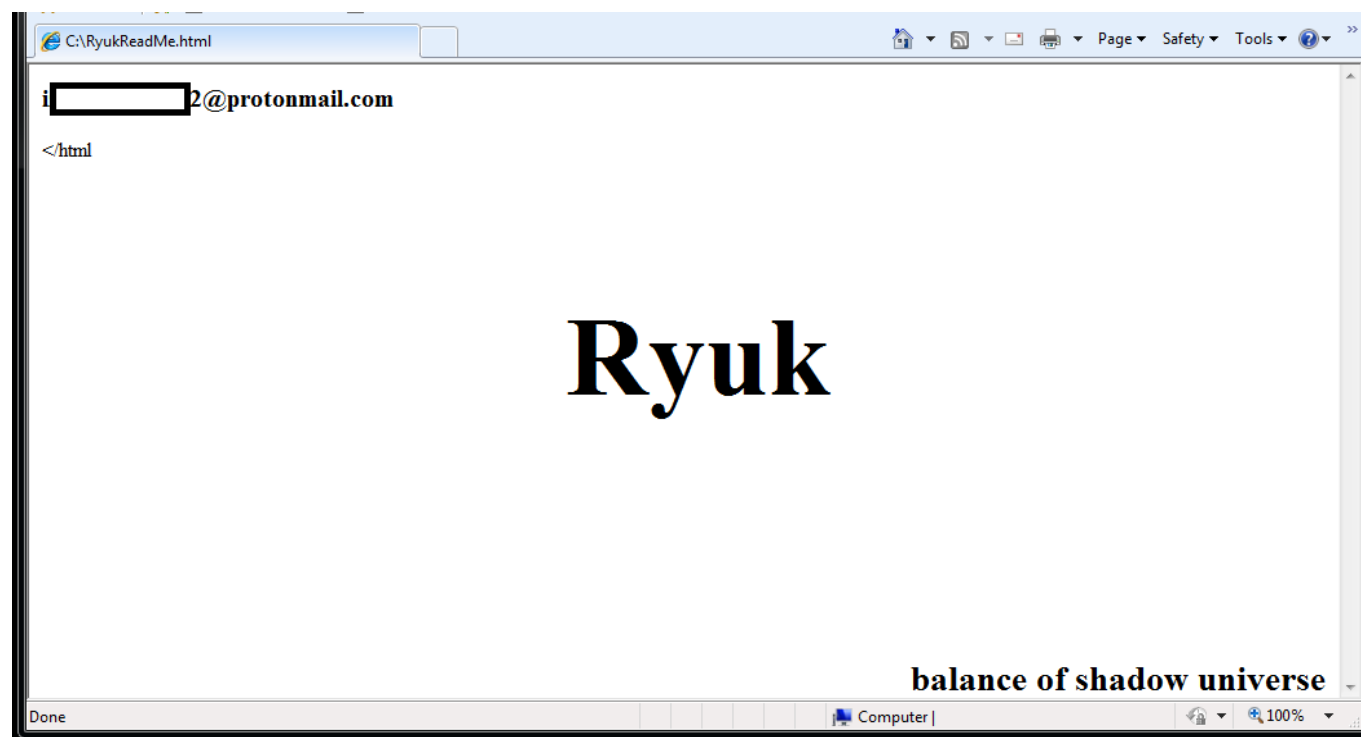


Figure: Ransom-Ryuk Ransom Note

## System Backup Deletion

The ransomware creates a Windows BAT file named: "window.bat" (location depends on the OS version) to create list of commands to delete backup shadow copies and files. Commands executed:

```
vssadmin Delete Shadows /all /quiet
vssadmin resize shadowstorage /for=c: /on=c: /maxsize=401MB
vssadmin resize shadowstorage /for=c: /on=c: /maxsize=unbounded
vssadmin resize shadowstorage /for=d: /on=d: /maxsize=401MB
vssadmin resize shadowstorage /for=d: /on=d: /maxsize=unbounded
vssadmin resize shadowstorage /for=e: /on=e: /maxsize=401MB
vssadmin resize shadowstorage /for=e: /on=e: /maxsize=unbounded
vssadmin resize shadowstorage /for=f: /on=f: /maxsize=401MB
vssadmin resize shadowstorage /for=f: /on=f: /maxsize=unbounded
vssadmin resize shadowstorage /for=g: /on=g: /maxsize=401MB
vssadmin resize shadowstorage /for=g: /on=g: /maxsize=unbounded
vssadmin resize shadowstorage /for=h: /on=h: /maxsize=401MB
vssadmin resize shadowstorage /for=h: /on=h: /maxsize=unbounded
vssadmin Delete Shadows /all /quiet
del /s /f /q c:\*.VHD c:\*.bac c:\*.bak c:\*.wbcat c:\*.bkf c:\Backup*.
c:\backup*. c:\*.set c:\*.win c:\*.dsk
del /s /f /q d:\*.VHD d:\*.bac d:\*.bak d:\*.wbcat d:\*.bkf d:\Backup*.
d:\backup*. d:\*.set d:\*.win d:\*.dsk
del /s /f /q e:\*.VHD e:\*.bac e:\*.bak e:\*.wbcat e:\*.bkf e:\Backup*.
e:\backup*. e:\*.set e:\*.win e:\*.dsk
del /s /f /q f:\*.VHD f:\*.bac f:\*.bak f:\*.wbcat f:\*.bkf f:\Backup*.
f:\backup*. f:\*.set f:\*.win f:\*.dsk
del /s /f /q g:\*.VHD g:\*.bac g:\*.bak g:\*.wbcat g:\*.bkf g:\Backup*.
g:\backup*. g:\*.set g:\*.win g:\*.dsk
del /s /f /q h:\*.VHD h:\*.bac h:\*.bak h:\*.wbcat h:\*.bkf h:\Backup*.
h:\backup*. h:\*.set h:\*.win h:\*.dsk
```

## Services Stopped on the System

Ransom-Ryuk stops the following services on the endpoint prior to starting its encryption process. These services may be stopped to serve two purposes:

- Stop user/application services on the system to enable successful encryption of their files on disk.
- Stop AV services to disable behavior-based protection services.

Services stopped on the endpoint:

```
stop "Acronis VSS Provider" /y
stop "Enterprise Client Service" /y
stop "Sophos Agent" /y
stop "Sophos AutoUpdate Service" /y
stop "Sophos Clean Service" /y
stop "Sophos Device Control Service" /y
stop "Sophos File Scanner Service" /y
stop "Sophos Health Service" /y
stop "Sophos MCS Agent" /y
stop "Sophos MCS Client" /y
stop "Sophos Message Router" /y
stop "Sophos Safestore Service" /y
stop "Sophos System Protection Service" /y
stop "Sophos Web Control Service" /y
stop "SQLsafe Backup Service" /y
stop "SQLsafe Filter Service" /y
stop "Symantec System Recovery" /y
stop "Veeam Backup Catalog Data Service" /y
stop AcronisAgent /y
stop AcrSch2Svc /y
stop Antivirus /y
stop ARSM /y
stop BackupExecAgentAccelerator /y
```



```
stop BackupExecAgentBrowser /y
stop BackupExecDeviceMediaService /y
stop BackupExecJobEngine /y
stop BackupExecManagementService /y
stop BackupExecRPCService /y
stop BackupExecVSSProvider /y
stop bedbg /y
stop DCAgent /y
stop EPSecurityService /y
stop EPUUpdateService /y
stop EraserSvc11710 /y
stop EsgShKernel /y
stop FA Scheduler /y
stop IISAdmin /y
stop IMAP4Svc /y
stop macmnsvc /y
stop masvc /y
stop MBAMService /y
stop MBEndpointAgent /y
stop McAfeeEngineService /y
stop McAfeeFramework /y
stop McAfeeFrameworkMcAfeeFramework /y
stop McShield /y
stop McTaskManager /y
stop mfemms /y
stop mfevtp /y
stop MMS /y
stop mozyprobackup /y
stop MsDtsServer /y
stop MsDtsServer100 /y
stop MsDtsServer110 /y
stop MSExchangeES /y
stop MSExchangeIS /y
stop MSExchangeMGMT /y
stop MSExchangeMTA /y
stop MSExchangeSA /y
stop MSExchangeSRS /y
stop MSOLAP$SQL_2008 /y
stop MSOLAP$SYSTEM_BGC /y
stop MSOLAP$TPS /y
stop MSOLAP$TPSAMA /y
stop MSSQL$BKUPEXEC /y
stop MSSQL$ECWDB2 /y
stop MSSQL$PRACTICEMGT /y
stop MSSQL$PRACTTICEBGC /y
stop MSSQL$PROFXENGAGEMENT /y
stop MSSQL$SBSMONITORING /y
stop MSSQL$SHAREPOINT /y
stop MSSQL$SQL_2008 /y
stop MSSQL$SYSTEM_BGC /y
stop MSSQL$TPS /y
stop MSSQL$TPSAMA /y
stop MSSQL$VEEAMSQL2008R2 /y
stop MSSQL$VEEAMSQL2012 /y
stop MSSQLFDLauncher /y
stop MSSQLFDLauncher$PROFXENGAGEMENT /y
stop MSSQLFDLauncher$SBSMONITORING /y
stop MSSQLFDLauncher$SHAREPOINT /y
stop MSSQLFDLauncher$SQL_2008 /y
stop MSSQLFDLauncher$SYSTEM_BGC /y
stop MSSQLFDLauncher$TPS /y
stop MSSQLFDLauncher$TPSAMA /y
stop MSSQLSERVER /y
stop MSSQLServerADHelper100 /y
stop MSSQLServerOLAPService /y
stop MySQL80 /y
stop MySQL57 /y
stop ntrtscan /y
stop OracleClientCache80 /y
stop PDVFSService /y
```

```
stop POP3Svc /y
stop ReportServer /y
stop ReportServer$SQL 2008 /y
stop ReportServer$SYSTEM_BGC /y
stop ReportServer$TPS /y
stop ReportServer$TPSAMA /y
stop RESvc /y
stop sacsvr /y
stop SamSs /y
stop SAVAdminService /y
stop SAVService /y
stop SDRSVC /y
stop SepMasterService /y
stop ShMonitor /y
stop Smcinst /y
stop SmcService /y
stop SMTPSvc /y
stop SNAC /y
stop SntpService /y
stop sophossps /y
stop SQLAgent$BKUPEXEC /y
stop SQLAgent$ECWDB2 /y
stop SQLAgent$PRACTTICEBGC /y
stop SQLAgent$PRACTTICEMGT /y
stop SQLAgent$PROFXENGAGEMENT /y
stop SQLAgent$SBSMONITORING /y
stop SQLAgent$SHAREPOINT /y
stop SQLAgent$SQL_2008 /y
stop SQLAgent$SYSTEM_BGC /y
stop SQLAgent$TPS /y
stop SQLAgent$TPSAMA /y
stop SQLAgent$VEEAMSQL2008R2 /y
stop SQLAgent$VEEAMSQL2012 /y
stop SQLBrowser /y
stop SQLSafeOLRService /y
stop SQLSERVERAGENT /y
stop SQLTELEMETRY /y
stop SQLTELEMETRY$ECWDB2 /y
stop SQLWriter /y
stop SstpSvc /y
stop svcGenericHost /y
stop swi_filter /y
stop swi_service /y
stop swi_update_64 /y
stop TmCCSF /y
stop tmlisten /y
stop TrueKey /y
stop TrueKeyScheduler /y
stop TrueKeyServiceHelper /y
stop UI0Detect /y
stop VeeamBackupSvc /y
stop VeeamBrokerSvc /y
stop VeeamCatalogSvc /y
stop VeeamCloudSvc /y
stop VeeamDeploymentService /y
stop VeeamDeploySvc /y
stop VeeamEnterpriseManagerSvc /y
stop VeeamMountSvc /y
stop VeeamNFSSvc /y
stop VeeamRESTSvc /y
stop VeeamTransportSvc /y
stop W3Svc /y
stop wbengine /y
stop WRSVC /y
stop MSSQL$VEEAMSQL2008R2 /y
stop SQLAgent$VEEAMSQL2008R2 /y
stop VeeamHvIntegrationSvc /y
stop swi_update /y
stop SQLAgent$CXDB /y
stop SQLAgent$CITRIX_METAFRAME /y
```

```

stop "SQL Backups" /y
stop MSSQL$PROD /y
stop "Zoolz 2 Service" /y
stop MSSQLServerADHelper /y
stop SQLAgent$PROD /y
stop msftesql$PROD /y
stop NetMsmqActivator /y
stop EhttpSrv /y
stop ekrn /y
stop ESHASRV /y
stop MSSQL$SOPHOS /y
stop SQLAgent$SOPHOS /y
stop AVP /y
stop klnagent /y
stop MSSQL$SQLEXPRESS /y
stop SQLAgent$SQLEXPRESS /y
stop wbengine /y
stop kavfsslp /y
stop KAVFSGT /y
stop KAVFS /y
stop mfefire /y

```

### Miscellaneous Functionality

- The second-stage binary also creates the following files on disk:

For Windows 7 and above: C:\users\Public\sys

For Windows XP/Windows 2000: C:\Documents and Settings\Default User\sys

For Windows 7 and above: C:\users\Public\finish

For Windows XP/Windows 2000: C:\Documents and Settings\Default User\finish

### WOL (Wake-On-Lan)

RYUK also spawns a child process, which is a self-copy of itself but with a command line argument **"8 LAN"**.

When RYUK is executed with this command line argument, it tries to wake up all the machines on the network by sending WOL (Wake-On-Lan) packets to all the machines, so that these machines can also be encrypted.

The image below shows the child process executed with **"8 LAN"** command line argument:

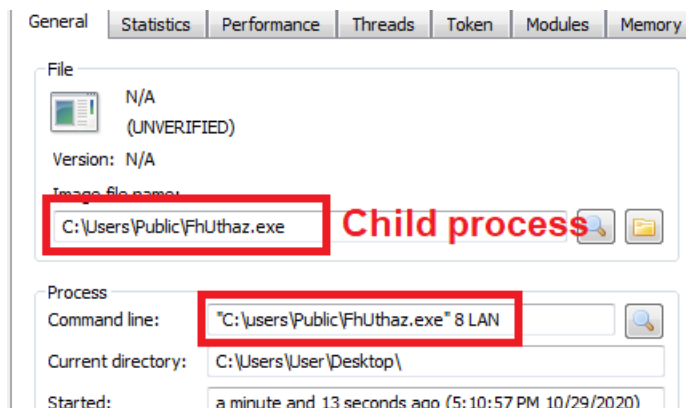


Figure: Child process with 8 LAN CLA.

The network capture below shows the WOL packets being broadcasted on the network on all the present IP ranges:

Wireshark - Packet 199 - Local Area Connection

Frame 199: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: Dell, Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Address: Broadcast (ff:ff:ff:ff:ff:ff)

... .. = LG bit: Locally administered address (this is NOT the factory default)

... ..1 = IG bit: Group address (multicast/broadcast)

Source: Dell

Type: ARP (0x0806)

Padding: 00000000000000000000000000000000

Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 78 2b cb 15 57 f2 08 06 00 01

0010 08 00 06 04 00 01 78 2b cb 15 57 f2 0a d5 e0 22

0020 00 00 00 00 00 00 01 d5 e0 15 00 00 00 00 00

0030 00 00 00 00 00 00 00 00 00 00 00 00

**WOL packet**

Figure: Child process with 8 LAN CLA.

## Restart Mechanism

The following registry entry would enable the malware to execute every time when Windows starts:

- The second-stage binary set up persistence on the system by executing the following command:

```
C:\Windows\system32\cmd.exe /C REG ADD
```

```
"HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /v "svchos"  
/t REG_SZ /d "<Path_to_2nd_stage_binary>" /f
```

The value of the Run key consists of the path of the second-stage binary.

## Indicators of compromise

McAfee has observed the following detections occurring in environments prior to a Ransom Ryuk Attack:

- PS/Dropper.b
- Trojan-FSMT
- Trojan-FQRU
- PS/Injector.p
- Cobalt-EVTS
- CobaltStr-FDWE
- Cobalt/Shell
- CobaltStrike
- CobaltStr
- Trojan-Cobalt
- AMSI-FHR

Although this is not an exhaustive list, if any of the above detections are observed in your environment – they may be indicators of attempted compromise by the Ryuk group. We therefore encourage customers to implement our recommended mitigation steps in their environment and conduct a full ODS scan.

## Getting Help from the McAfee Foundstone Services team

This document is intended to provide a summary of current intelligence and best practices to ensure the highest level of protection from your McAfee security solution. The McAfee Foundstone Services team offers a full range of strategic and technical consulting services that can further help to ensure you identify security risk and build effective solutions to remediate security vulnerabilities.

You can reach them here: <https://www.mcafee.com/enterprise/en-us/services/foundstone-services.html>

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